

Latent Bias Correction in Outpainting Artworks

Jung-Jae Yu*

Content Research Division, Electronics and
Telecommunications Research Institute (ETRI)
South Korea
jungjae@etri.re.kr

Dae-Young Song*

Content Research Division, Electronics and
Telecommunications Research Institute (ETRI)
South Korea
eadyoung@etri.re.kr



Figure 1: Comparison of art painting results of vanilla/previous/proposed outpainting method

Abstract

This paper describes research on applying latent correction to image outpainting using a diffusion model. The purpose of this study is to eliminate unnecessary tendencies that frequently occur when outpainting an artwork, which lower the completeness of the resulting work. The core of the proposed method is to apply the error correction value calculated in the already known input region to the area to be generated during the denoising process in the latent space. The effectiveness of the proposed method was verified quantitatively and qualitatively through experiments.

CCS Concepts

• **Applied computing** → *Media arts*; • **Computing methodologies** → *Computer vision representations*; *Image manipulation*.

*Both authors contributed equally to this research.

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SA Posters '24, December 03–06, 2024, Tokyo, Japan
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ACM ISBN 979-8-4007-1138-1/24/12
<https://doi.org/10.1145/3681756.3697914>

Keywords

Diffusion, Image, Outpainting, Art, Latent, Correction

ACM Reference Format:

Jung-Jae Yu and Dae-Young Song. 2024. Latent Bias Correction in Outpainting Artworks. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697914>

1 Introduction

Recently, the quality of image generation has been greatly improved by the use of the diffusion method. A good result in outpainting an artwork requires generating an outer region that is naturally connected to the content provided in the input image region while maintaining the same color tone and style. Some of the problems that frequently occur under such criteria are as follows: first, water elements that were not in the original image are generated in the lower region (Figure 1-(a)); second, the original image is expressed as if it were contained in an artificial frame (Figure 1-(b)); third, the color of the extended area is not consistent with the input image, or additional unnatural objects are added (Figure 1-(c)). To solve these problems, we developed a very simple yet effective method to improve the quality of the generated outer region by utilizing information from the input image region in the latent space, and verified its effectiveness through experiments.

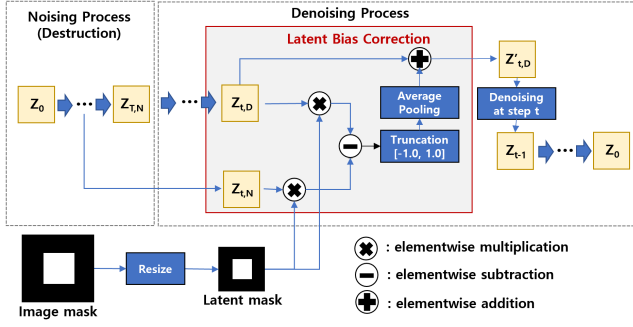


Figure 2: Proposed method for latent bias correction

2 Related Works

Since the diffusion method requires much computation, Rombach et al. proposed the denoising process in the latent space instead of the pixel space [Rombach et al. 2022a]. Avrahami et al. proposed a latent blending for Latent Diffusion Model(LDM)-based inpainting task [Avrahami et al. 2023]. For region to be preserved, They proposed to overwrite latent code obtained during the diffusion process into the corresponding area during the denoising process, and showed that this makes the result better. However we observed that latent blending alone does not sufficiently resolve the frequently occurring artifacts described in the introduction.

3 Approach

The proposed method is inspired by the following intuition. At each step of the denoising process, the latent features in the outer region can be compensated by using the estimation error calculated in the central region. This approach is possible because it is expected that undesirable tendencies in the denoising process may have some degree of interdependence between the central region and the outer region. Although the positional relationship between the image pixel and the latent space does not correspond exactly, Avrahami et al. have already shown that it is meaningful to apply the same masking in the latent space [Avrahami et al. 2023].

Figure 2 describes our method for bias correction in the latent space. $Z_{t,N}$ is a latent code in the diffusion process that adds noise, and $Z_{t,D}$ is a latent code in the denoising process. Before estimating the noise at each step, we calculate the average value of $Z_{t,N} - Z_{t,D}$ in the central region, and add this residual to $Z_{t,D}$ of the entire region including the outer area to be generated. In this process, truncation is applied to reflect the overall tendency without relying excessively on some restricted region. Denoising at each step is performed with the corrected $Z'_{t,D}$.

This method was applied to the previous inpainting [Rombach et al. 2022b] framework with 25 step DDIM. Existing latent blending [Avrahami et al. 2023] is also used with the proposed bias correction. Considering that the pixel mask does not exactly match the latent mask and that the overall composition of the image is determined in the early steps, both latent blending and latent bias correction are applied only in the first T/4 steps. The text prompts for outpainting were generated by inputting the uncropped artworks into BLIP [Li et al. 2022].

Table 1: Number of occurrences of each bias type from 1,267 test images and successful bias correction rate

Unnatural Bias type	Biased Images	Corrected Images	Correction Ratio
Water at bottom	117	99	84.6%
Artificial frame	168	134	79.8%
Style Mismatch	166	156	94.0%

4 Experiments

1,267 images in various styles including Classicism, Baroque, and Rococo were sampled from the WikiArt dataset. As shown in Figure 1, we conduct experiments under various conditions, while simultaneously setting the expansion ratio to twofold, creating a challenging experimental environment.

Table 1 shows the frequency with which the undesirable tendencies described in introduction occurred and the rate of successful correction with the proposed method. We found that for each bias type, approximately 80% or more were resolved. Table 1 is the result obtained by visually confirming the results of outpainting, and the result images are submitted as supplementary material.

For quantitative evaluation, we compute frechet inception distance (FID) between the results and the originals, where a lower FID score indicates better generation quality. When the pretrained model [Rombach et al. 2022b] is applied alone, the score is 40.5. With the addition of latent blending [Avrahami et al. 2023], the score increases to 42.1. However, by incorporating our method, we observed an improvement in quality, with the FID score reduced to 39.6.

5 Conclusion and Discussion

We have presented a method to correct the undesirable tendency that frequently occurs when outpainting artworks, and verified its usefulness through experiments without additional training. Of course, it cannot be said that this treatment alone always produces high-quality results in outpainting works of art, but the significance of this study lies in the fact that it solves several types of problems that frequently occur.

Acknowledgments

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (No. RS-2024-00395401, Development of VFX creation and combination using generative AI)

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